

Raw data:

| bcr_patient_barcode |                       | tcga-05-4245                         | tcga-05-4382                         | tcga-05-4384                         | tcga-05-4396                         | tcga-05-4402                         | tcga-05-4405                         | tcga-05-4410                         | tcga-05-4418                         | tcga-05-4427                         | ... |
|---------------------|-----------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|-----|
| 0                   | bcr_patient_uid       | 03d09c05-49ab-4ba6-a8d7-e7cc7f1faf2d | 3434b91a-c05f-460f-a078-7b1bb6e7085d | 9a50e7e4-831d-489f-87d2-979e987561cc | 9f81c602-8afa-4588-b0b6-6e5a1a128d5a | 722d6818-f1cb-49f4-b68a-fd0a31304681 | f587c9ab-2949-4410-80d6-a050865d48aa | d28e465c-e553-4c53-8bb7-3d8199b9a8a5 | d2c1e896-6886-4122-bb48-5fbc3f641f14 | 78a24c56-576c-4245-bb42-6603b3d19897 | ... |
| 1                   | patient_id            | 4245                                 | 4382                                 | 4384                                 | 4396                                 | 4402                                 | 4405                                 | 4410                                 | 4418                                 | 4427                                 | ... |
| 2                   | vital_status          | alive                                | alive                                | alive                                | dead                                 | dead                                 | alive                                | alive                                | dead                                 | alive                                | ... |
| 3                   | days_to_death         | NaN                                  | NaN                                  | NaN                                  | 303                                  | 244                                  | NaN                                  | NaN                                  | 274                                  | NaN                                  | ... |
| 4                   | days_to_last_followup | 730                                  | 607                                  | 426                                  | NaN                                  | NaN                                  | 610                                  | 0                                    | NaN                                  | 791                                  | ... |

5 rows x 523 columns

Transpose the dataset and filter for desired stages.

| bcr_patient_barcode | bcr_patient_uid                      | patient_id | vital_status | days_to_death | days_to_last_followup | additional_studies | age_at_initial_pathologic_diagnosis | anator |
|---------------------|--------------------------------------|------------|--------------|---------------|-----------------------|--------------------|-------------------------------------|--------|
| tcga-05-4245        | 03d09c05-49ab-4ba6-a8d7-e7ccf71fafd2 | 4245       | alive        | NaN           | 730                   | NaN                |                                     | 81     |
| tcga-05-4382        | 3434b91a-c05f-460f-a078-7b1bb6e7085d | 4382       | alive        | NaN           | 607                   | NaN                |                                     | 68     |
| tcga-05-4384        | 9a50e7e4-831d-489f-87d2-979e987561cc | 4384       | alive        | NaN           | 426                   | NaN                |                                     | 66     |
| tcga-05-4396        | 9f81c602-8afa-4588-b0b6-6e5a1a128d5a | 4396       | dead         | 303           | NaN                   | NaN                |                                     | 76     |
| tcga-05-4402        | 722d6818-f1cb-49f4-b68a-fd0a31304681 | 4402       | dead         | 244           | NaN                   | NaN                |                                     | 57     |

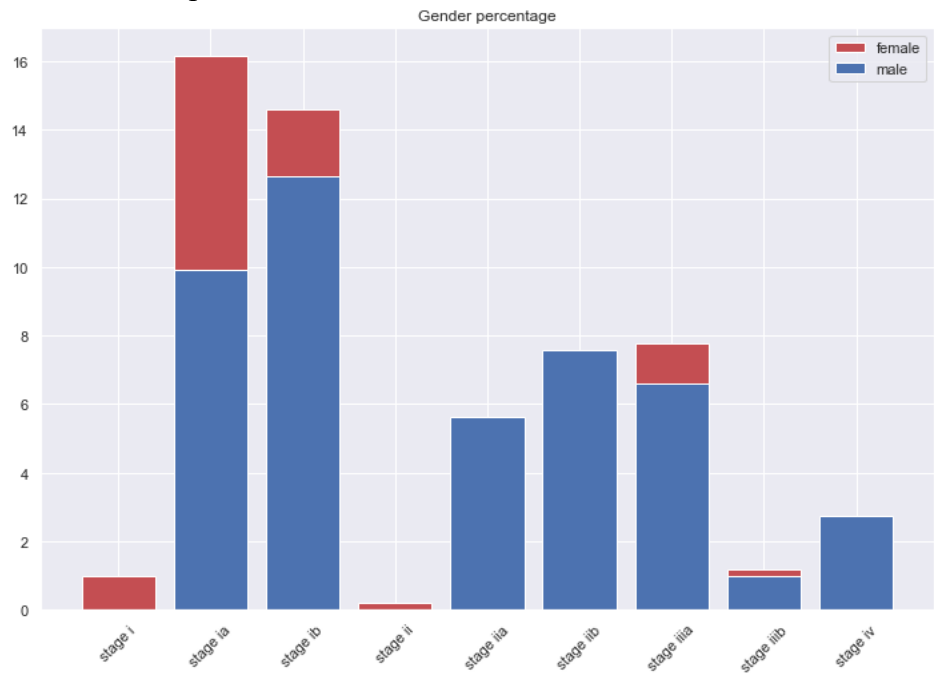
5 rows x 102 columns

```
['stage i',
 'stage ia',
 'stage ib',
 'stage ii',
 'stage iia',
 'stage iib',
 'stage iiia',
 'stage iiib',
 'stage iv']
```

Proportion of each stage by gender:

|    | pathologic_stage | gender | count | Percentage |
|----|------------------|--------|-------|------------|
| 0  | stage i          | female | 5     | 0.972763   |
| 1  | stage ia         | female | 83    | 16.147860  |
| 2  | stage ia         | male   | 51    | 9.922179   |
| 3  | stage ib         | female | 75    | 14.591440  |
| 4  | stage ib         | male   | 65    | 12.645914  |
| 5  | stage ii         | female | 1     | 0.194553   |
| 6  | stage iia        | female | 21    | 4.085603   |
| 7  | stage iia        | male   | 29    | 5.642023   |
| 8  | stage iib        | female | 34    | 6.614786   |
| 9  | stage iib        | male   | 39    | 7.587549   |
| 10 | stage iiaa       | female | 40    | 7.782101   |
| 11 | stage iiaa       | male   | 34    | 6.614786   |
| 12 | stage iiib       | female | 6     | 1.167315   |
| 13 | stage iiib       | male   | 5     | 0.972763   |
| 14 | stage iv         | female | 12    | 2.334630   |
| 15 | stage iv         | male   | 14    | 2.723735   |

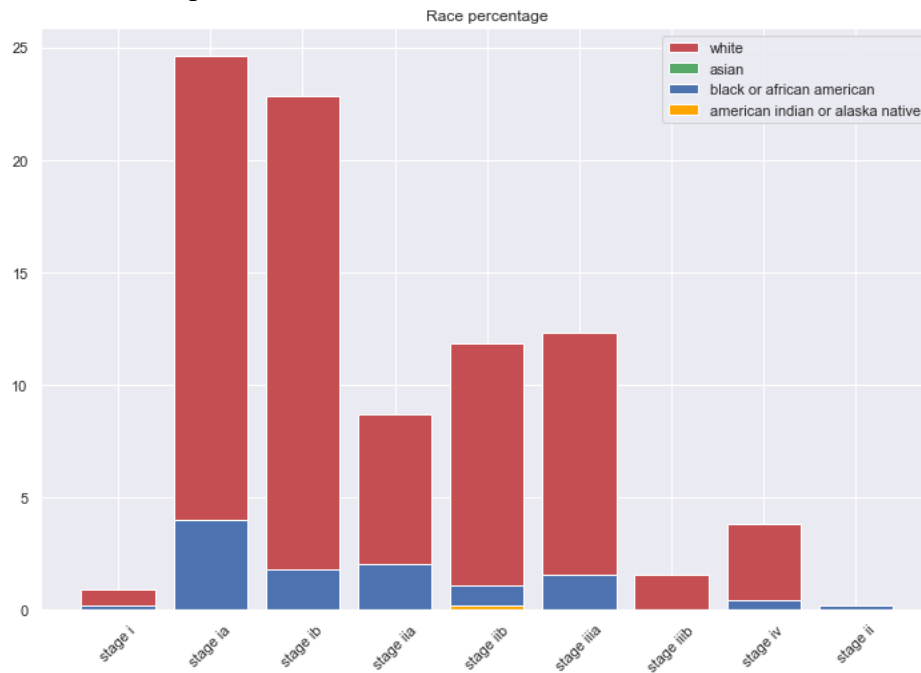
In stacked bar plot:



Proportion of each stage by age:

|    | pathologic_stage | race                             | count | Percentage |
|----|------------------|----------------------------------|-------|------------|
| 0  | stage i          | black or african american        | 1     | 0.223714   |
| 1  | stage i          | white                            | 4     | 0.894855   |
| 2  | stage ia         | asian                            | 2     | 0.447427   |
| 3  | stage ia         | black or african american        | 18    | 4.026846   |
| 4  | stage ia         | white                            | 110   | 24.608501  |
| 5  | stage ib         | asian                            | 3     | 0.671141   |
| 6  | stage ib         | black or african american        | 8     | 1.789709   |
| 7  | stage ib         | white                            | 102   | 22.818792  |
| 8  | stage ii         | black or african american        | 1     | 0.223714   |
| 9  | stage iia        | black or african american        | 9     | 2.013423   |
| 10 | stage iia        | white                            | 39    | 8.724832   |
| 11 | stage iib        | american indian or alaska native | 1     | 0.223714   |
| 12 | stage iib        | asian                            | 1     | 0.223714   |
| 13 | stage iib        | black or african american        | 5     | 1.118568   |
| 14 | stage iib        | white                            | 53    | 11.856823  |
| 15 | stage iia        | black or african american        | 7     | 1.565996   |
| 16 | stage iia        | white                            | 55    | 12.304251  |
| 17 | stage iib        | white                            | 7     | 1.565996   |
| 18 | stage iv         | asian                            | 2     | 0.447427   |
| 19 | stage iv         | black or african american        | 2     | 0.447427   |
| 20 | stage iv         | white                            | 17    | 3.803132   |

In stacked bar plot:



Proportion of each stage by both gender and age:

|    | pathologic_stage | gender | race                             | count | Percentage |
|----|------------------|--------|----------------------------------|-------|------------|
| 0  | stage i          | female | black or african american        | 1     | 0.22371365 |
| 1  | stage i          | female | white                            | 4     | 0.89485459 |
| 2  | stage ia         | female | asian                            | 2     | 0.44742729 |
| 3  | stage ia         | female | black or african american        | 14    | 3.13199105 |
| 4  | stage ia         | female | white                            | 64    | 14.3176734 |
| 5  | stage ia         | male   | black or african american        | 4     | 0.89485459 |
| 6  | stage ia         | male   | white                            | 46    | 10.2908277 |
| 7  | stage ib         | female | asian                            | 2     | 0.44742729 |
| 8  | stage ib         | female | black or african american        | 3     | 0.67114094 |
| 9  | stage ib         | female | white                            | 60    | 13.4228188 |
| 10 | stage ib         | male   | asian                            | 1     | 0.22371365 |
| 11 | stage ib         | male   | black or african american        | 5     | 1.11856823 |
| 12 | stage ib         | male   | white                            | 42    | 9.39597315 |
| 13 | stage ii         | female | black or african american        | 1     | 0.22371365 |
| 14 | stage iia        | female | black or african american        | 3     | 0.67114094 |
| 15 | stage iia        | female | white                            | 18    | 4.02684564 |
| 16 | stage iia        | male   | black or african american        | 6     | 1.34228188 |
| 17 | stage iia        | male   | white                            | 21    | 4.69798658 |
| 18 | stage iib        | female | american indian or alaska native | 1     | 0.22371365 |
| 19 | stage iib        | female | black or african american        | 2     | 0.44742729 |
| 20 | stage iib        | female | white                            | 26    | 5.81655481 |
| 21 | stage iib        | male   | asian                            | 1     | 0.22371365 |

|    |            |        |                           |    |            |
|----|------------|--------|---------------------------|----|------------|
| 22 | stage iib  | male   | black or african american | 3  | 0.67114094 |
| 23 | stage iib  | male   | white                     | 27 | 6.04026846 |
| 24 | stage iiia | female | black or african american | 4  | 0.89485459 |
| 25 | stage iiia | female | white                     | 31 | 6.93512304 |
| 26 | stage iiia | male   | black or african american | 3  | 0.67114094 |
| 27 | stage iiia | male   | white                     | 24 | 5.36912752 |
| 28 | stage iiib | female | white                     | 5  | 1.11856823 |
| 29 | stage iiib | male   | white                     | 2  | 0.44742729 |
| 30 | stage iv   | female | black or african american | 1  | 0.22371365 |
| 31 | stage iv   | female | white                     | 7  | 1.56599553 |
| 32 | stage iv   | male   | asian                     | 2  | 0.44742729 |
| 33 | stage iv   | male   | black or african american | 1  | 0.22371365 |
| 34 | stage iv   | male   | white                     | 10 | 2.23713647 |

In stacked bar plot:

